

The Deterministic CBigNum v1.4 Serialization Fingerprint Across Early Bitcoin Coinbase Payloads

Lead Researcher: Qahtan Al-Oud
Email: qahtanbins1@gmail.com

Research Paper Framework

A structured technical and forensic research framework for reproducible analysis of early Bitcoin mainnet coinbase payloads.

1. Title

The Deterministic CBigNum v1.4 Serialization Fingerprint Across Early Bitcoin Coinbase Payloads

2. Abstract

“We identify and reproducibly validate a deterministic CBigNum v1.4 serialization fingerprint across early Bitcoin coinbase payloads, with 8,713+ exact matches across the first 10,000 mainnet blocks and comparative testing against alternative encoding models.”

This research presents a forensic audit of Bitcoin mainnet blocks 1 to 10,000 under the frozen v1.4 protocol, resolving 9,922 conforming blocks and 78 boundary anomalies driven by CBigNum::getvch() sign-bit shift mechanics.

3. Introduction & Scope

Scope: Mainnet blocks 1 to 10,000 under protocol version v1.4 Frozen.

Core Objective: Establishing a reproducible forensic analysis of early coinbase ExtraNonce structures.

4. Methodology & Execution Audit

Total Audited Blocks: 10,000 blocks.

Conformance Results: 9,922 conforming records, 78 boundary anomalies, and 0 parsing errors.

Integrity Proof: Finalized SHA-256 cryptographic hashes for research packages and execution audit trails (binance_research_package.json).

5. Alternative-Encoding & Null Model Challenge

Comparative Models Tested: CBigNum Historical (v1.4), CScriptNum (Signed LE), Unsigned LE, and Random Null Baselines.

Empirical Findings: Demonstration of exclusive 8,713+ exact matches for CBigNum while alternative models experience divergence at sign-bit shift boundaries.

6. On-Chain Verifier Architecture

Implementation of verification smart contract ABI featuring verifyBlockConformance and checkBoundaryAnomaly functions to support decentralized reproducibility.

Publication Positioning

The framework presents the serialization finding as a reproducible technical and forensic result. A serialization fingerprint alone should be distinguished from broader claims about the identity of a particular miner or individual.

Reproducibility Note

The numerical findings and terminology in this document are reproduced from the supplied research framework and should be independently verified against the underlying benchmark artifacts and raw data before being characterized as definitive historical proof.